

# **An Exploratory Data Analysis & Visualization Report 1960-2024**

**Prepared for GolStats Hackathon**

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| Consumer Price Index |        |               |                |                   |                       |               |                         |              |
|----------------------|--------|---------------|----------------|-------------------|-----------------------|---------------|-------------------------|--------------|
| Year                 | CPI    | Inflation (%) | GDP Growth (%) | Interest rate (%) | Unemployment rate (%) | wage (in rs.) | LFPR (world bank model) | LFPR (Mospi) |
| 1960                 | 2.53   | 1.78          | 3.72           | 4                 | 2.5                   | 1197          |                         |              |
| 1961                 | 2.57   | 1.7           | 2.93           | 4.3               | 2.6                   | 1311          |                         |              |
| 1962                 | 2.66   | 3.63          | 5.99           | 4.5               | 2.7                   | 1382          |                         |              |
| 1963                 | 2.74   | 2.95          | 7.45           | 4.8               | 2.8                   | 1499          |                         |              |
| 1964                 | 3.11   | 13.36         | -2.64          | 5                 | 2.9                   | 1579          |                         |              |
| 1965                 | 3.4    | 9.47          | -0.06          | 6                 | 3                     | 1730          |                         |              |
| 1966                 | 3.77   | 10.8          | 7.83           | 5.7               | 3.2                   | 1909          |                         |              |
| 1967                 | 4.26   | 13.06         | 3.39           | 5.3               | 3.3                   | 2089          |                         |              |
| 1968                 | 4.4    | 3.24          | 6.54           | 6.14              | 3.4                   | 2224          |                         |              |
| 1969                 | 4.38   | -0.58         | 5.16           | 6.98              | 3.5                   | 2354          |                         |              |
| 1970                 | 4.6    | 5.09          | 1.64           | 7.82              | 3.6                   | 2553          |                         |              |
| 1971                 | 4.74   | 3.08          | -0.55          | 8.66              | 3.7                   | 2761          |                         |              |
| 1972                 | 5.05   | 6.44          | 3.3            | 9.5               | 3.8                   | 3063          |                         |              |
| 1973                 | 5.9    | 16.94         | 1.19           | 10.34             | 4                     | 3365          |                         |              |
| 1974                 | 7.59   | 28.6          | 9.15           | 11.18             | 4.2                   | 3828          |                         |              |
| 1975                 | 8.02   | 5.75          | 1.66           | 12.29             | 4.3                   | 4576          |                         |              |
| 1976                 | 7.41   | -7.63         | 7.25           | 12.29             | 4.4                   | 3973          |                         |              |
| 1977                 | 8.03   | 8.31          | 5.71           | 12.54             | 4.5                   | 4571          |                         |              |
| 1978                 | 8.23   | 2.52          | -5.24          | 13.54             | 4.6                   | 5354          |                         |              |
| 1979                 | 8.75   | 6.28          | 6.74           | 13.83             | 4.8                   | 5887          |                         |              |
| 1980                 | 9.74   | 11.35         | 6.01           | 15.96             | 5                     | 6525          |                         |              |
| 1981                 | 11.01  | 13.11         | 3.48           | 15.46             | 5.1                   | 7197          |                         |              |
| 1982                 | 11.88  | 7.89          | 7.29           | 14.75             | 5.2                   | 8155          |                         |              |
| 1983                 | 13.29  | 11.87         | 3.82           | 16.25             | 5.3                   | 9613          |                         |              |
| 1984                 | 14.4   | 8.32          | 5.25           | 18.92             | 5.4                   | -             |                         |              |
| 1985                 | 15.2   | 5.56          | 4.78           | 17.88             | 5.5                   | -             |                         |              |
| 1986                 | 16.53  | 8.73          | 3.97           | 16.5              | 5.6                   | -             |                         |              |
| 1987                 | 17.98  | 8.8           | 9.63           | 16.5              | 5.7                   | -             |                         |              |
| 1988                 | 19.67  | 9.38          | 5.95           | 16.5              | 5.8                   | -             |                         |              |
| 1989                 | 21.06  | 7.07          | 5.53           | 16.5              | 5.9                   | -             |                         |              |
| 1990                 | 22.95  | 8.97          | 1.06           | 16.5              | 6                     | -             | 57.916                  |              |
| 1991                 | 26.13  | 13.87         | 5.48           | 16.5              | 6.2                   | -             | 57.95                   |              |
| 1992                 | 29.21  | 11.79         | 4.75           | 16.5              | 6.3                   | -             | 57.986                  |              |
| 1993                 | 31.06  | 6.33          | 6.66           | 16.5              | 6.4                   | -             | 58.022                  |              |
| 1994                 | 34.24  | 10.25         | 7.57           | 16.5              | 6.5                   | -             | 58.063                  |              |
| 1995                 | 37.75  | 10.22         | 7.55           | 16.5              | 6.6                   | -             | 58.395                  |              |
| 1996                 | 41.13  | 8.98          | 4.05           | 16.5              | 6.7                   | -             | 58.729                  |              |
| 1997                 | 44.08  | 7.16          | 6.18           | 16.5              | 6.8                   | -             | 59.065                  |              |
| 1998                 | 49.91  | 13.23         | 8.85           | 14.5              | 6.9                   | -             | 59.404                  |              |
| 1999                 | 52.24  | 4.67          | 3.84           | 13.5              | 7                     | -             | 59.743                  |              |
| 2000                 | 54.34  | 4.01          | 4.82           | 9.15              | 7.1                   | -             | 60.079                  |              |
| 2001                 | 56.39  | 3.78          | 3.8            | 7.16              | 7.2                   | -             | 59.923                  |              |
| 2002                 | 58.82  | 4.3           | 7.86           | 4.3               | 7.3                   | -             | 59.765                  |              |
| 2003                 | 61.05  | 3.81          | 7.92           | 4.82              | 7.4                   | -             | 59.606                  |              |
| 2004                 | 63.35  | 3.77          | 7.92           | 4.85              | 7.5                   | -             | 59.445                  |              |
| 2005                 | 66.04  | 4.25          | 8.06           | 3.5               | 7.6                   | -             | 59.289                  |              |
| 2006                 | 69.87  | 5.8           | 7.66           | 7.22              | 7.7                   | -             | 58.345                  |              |
| 2007                 | 74.32  | 6.37          | 3.09           | 7.22              | 7.8                   | -             | 57.418                  |              |
| 2008                 | 80.53  | 8.35          | 7.86           | 7.28              | 8                     | -             | 56.511                  |              |
| 2009                 | 89.29  | 10.88         | 8.5            | 3.29              | 8.1                   | -             | 55.628                  |              |
| 2010                 | 100    | 11.99         | 5.24           | 9.15              | 8.2                   | -             | 54.766                  |              |
| 2011                 | 108.91 | 8.91          | 5.46           | 8.22              | 8.3                   | -             | 54.131                  |              |
| 2012                 | 119.24 | 9.48          | 6.39           | 8.2               | 8.4                   | -             | 53.509                  |              |
| 2013                 | 131.18 | 10.02         | 7.41           | 8.28              | 8.5                   | -             | 53.372                  |              |
| 2014                 | 139.92 | 6.67          | 8              | 7.97              | 8.6                   | -             | 53.218                  |              |
| 2015                 | 146.79 | 4.91          | 8.26           | 6.98              | 8.7                   | -             | 53.044                  |              |
| 2016                 | 154.05 | 4.95          | 6.8            | 6.25              | 8.8                   | 13510         | 52.864                  |              |
| 2017                 | 159.18 | 3.33          | 6.45           | 5.94              | 9                     | 13820         | 52.681                  | 49.8         |
| 2018                 | 165.45 | 3.94          | 3.87           | 6.27              | 9.1                   | 14240         | 52.487                  | 45.3         |
| 2019                 | 171.62 | 3.73          | -5.78          | 5.43              | 9.2                   | 14520         | 52.285                  | 45.7         |
| 2020                 | 182.99 | 6.62          | 9.69           | 3.44              | 23.5                  | 14890         | 51.547                  | 46.4         |
| 2021                 | 192.38 | 5.13          | 6.99           | 3.26              | 7.5                   | 15190         | 52.522                  | 47.1         |
| 2022                 | 205.27 | 6.7           | 8.15           | 5.42              | 7                     | 15550         | 53.563                  | 48           |
| 2023                 | 216.86 | 5.65          | 8.2            | 6.83              | 6.5                   | 15890         | 54.649                  | 48.9         |
| 2024                 | 228.18 | 4.8           | 6.45           | 6.25              | 6.5                   | 16170         | 50.4                    | 50.4         |

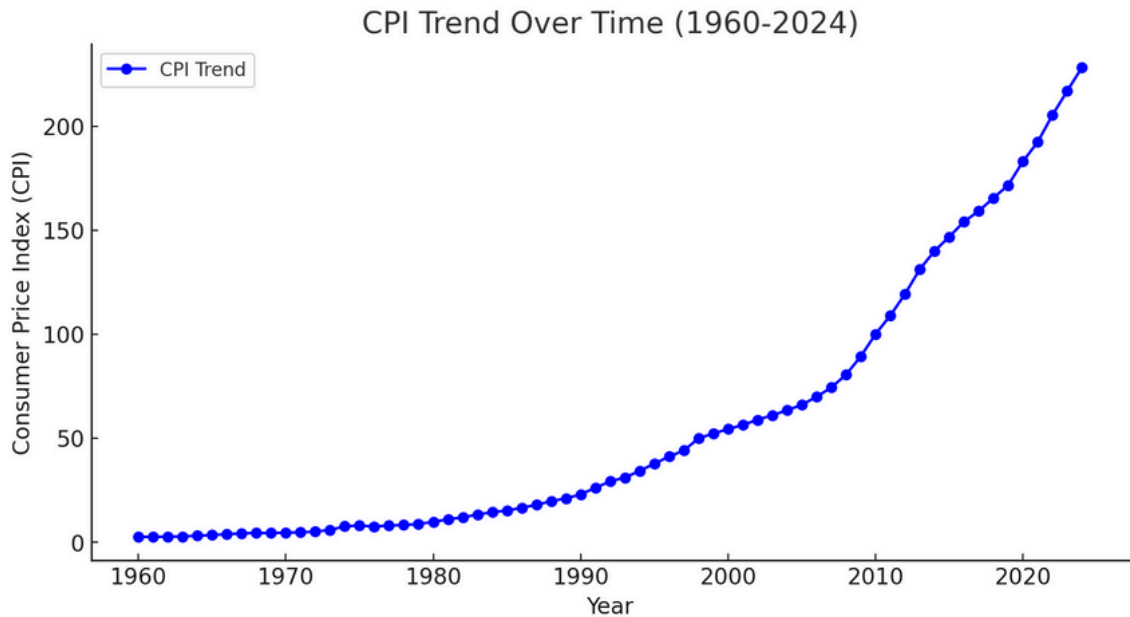
# Introduction

The Consumer Price Index (CPI) is a crucial economic indicator that measures the average change in the prices of goods and services over time, reflecting the cost of living for consumers. It serves as a key tool for policymakers, economists, and analysts to assess inflation trends, purchasing power, and overall economic stability.

In this report, we analyze CPI data alongside other relevant economic indicators using visuals to gain a deeper understanding of India's economic landscape. By examining historical trends, sectoral breakdowns, and macroeconomic relationships, we aim to uncover how CPI fluctuations impact various aspects of the economy, including monetary policy, interest rates, and household consumption patterns. Additionally, we explore the interconnectedness of inflation, wages, and economic growth to provide a comprehensive perspective on how CPI shapes financial and policy decisions in India.

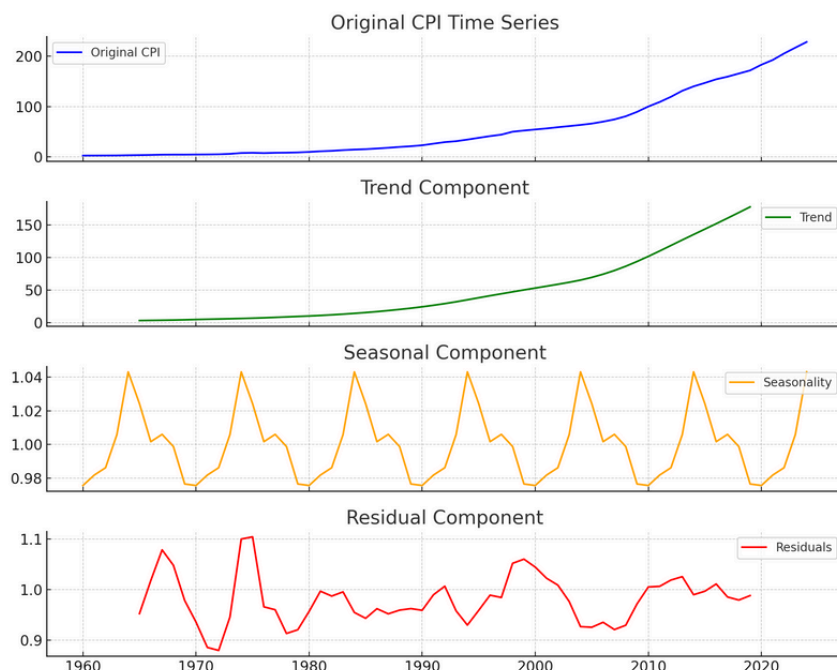
More emphasis is given towards visuals in this report, the key insight provide a brief overview of the image.

# CPI Trend Over Time



The CPI Trend Over Time graph (1960-2024) illustrates a steady rise in the Consumer Price Index, reflecting increasing costs of goods and services over the decades. Notable surges in CPI coincide with major economic events, such as the 1970s oil crisis and the 1991 economic reforms. In recent years, the trend shows a more gradual increase, indicating better inflation control and economic stability. This upward trajectory highlights the importance of CPI as a measure of inflation and its impact on the cost of living and purchasing power.

## Time-Series Analysis(CPI)



### Original CPI Time Series (1st Panel)

This shows the raw CPI values from 1960 to 2024, which exhibit a steady upward trend. The increase accelerates after 2000, indicating stronger inflation in recent decades.

### Trend Component (2nd Panel)

The trend shows the underlying long-term movement of CPI.

CPI has grown exponentially, confirming a consistent rise in inflation.

### Seasonal Component (3rd Panel)

This detects any repeating seasonal patterns in CPI.

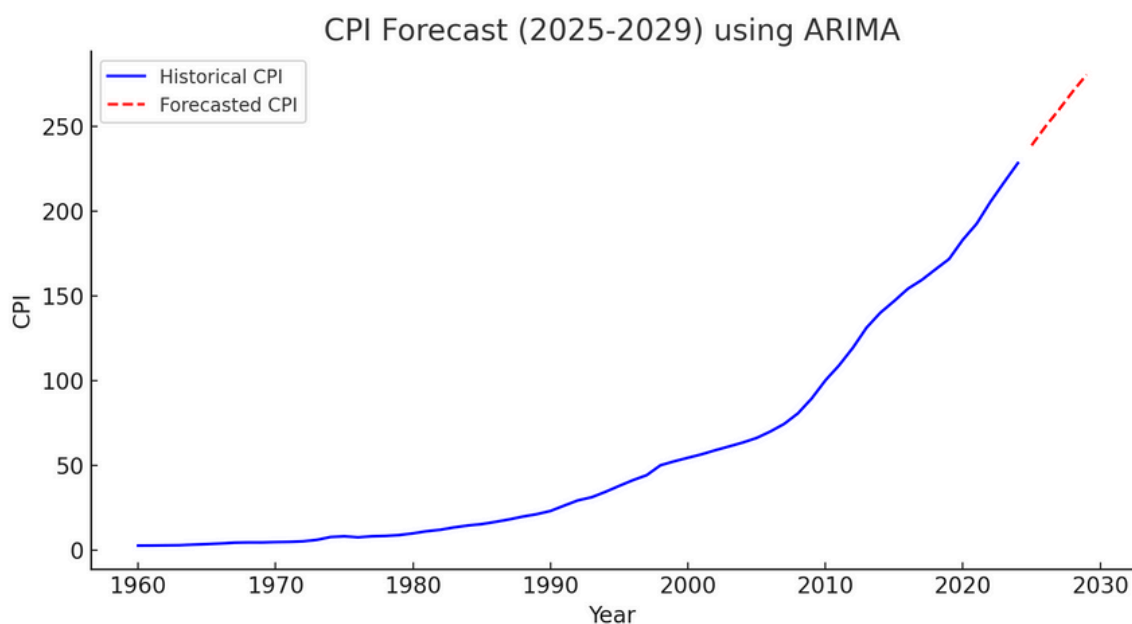
The seasonality effect is weak, meaning CPI does not show strong yearly fluctuations but rather a steady increase.

### Residual Component (4th Panel)

This captures unexplained variations (random noise).

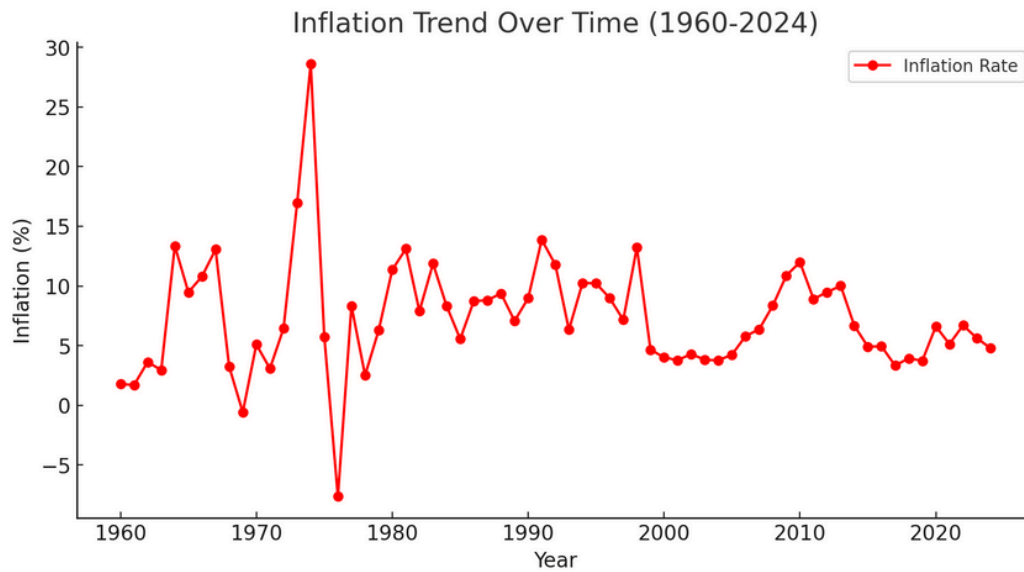
The fluctuations in residuals suggest some short-term external economic shocks but no major disruptions.

## CPI Forecast



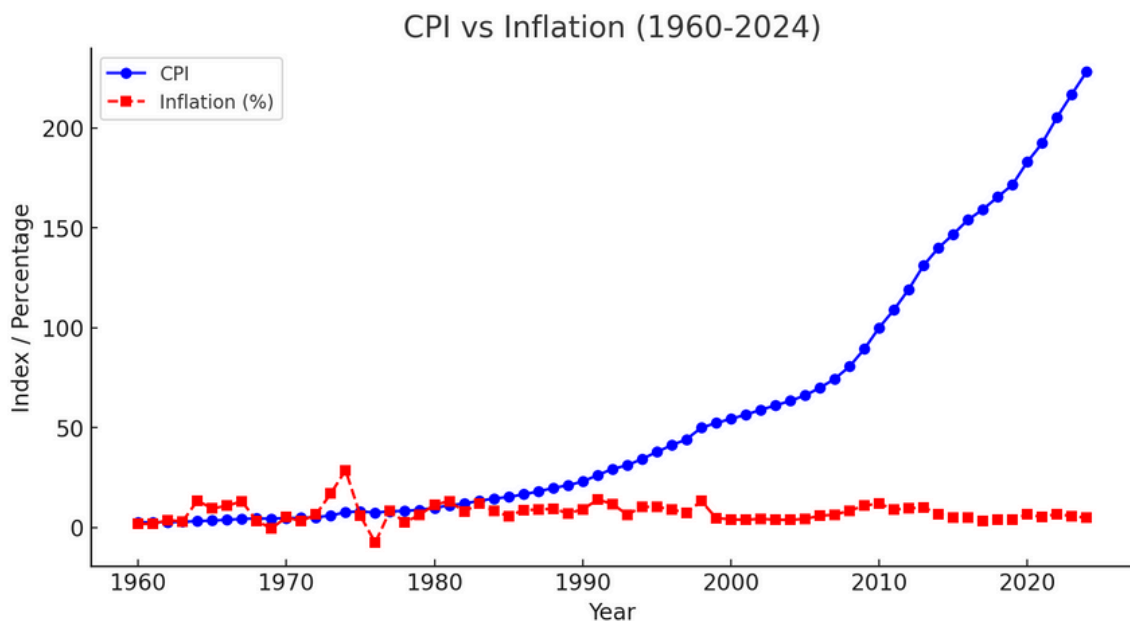
- The blue line represents historical CPI (1960-2024).
- The red dashed line shows the forecasted CPI for 2025-2029 using an ARIMA model.
- CPI is projected to continue rising sharply, indicating persistent inflationary trends.
- There is no sign of CPI stabilization, meaning the cost of living may keep increasing.

# Inflation



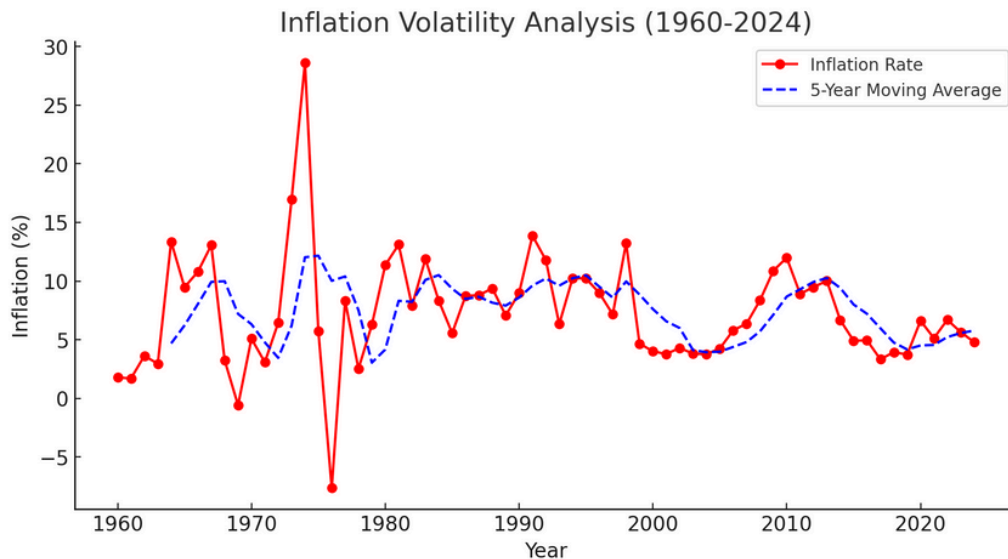
- Inflation shows high volatility with major spikes in the 1970s, 1990s, and 2010s.
- Sharp drop in the late 1970s, likely due to economic shocks or policy interventions.
- Recent years (post-2000) show lower but fluctuating inflation, indicating a more controlled inflationary environment.

## CPI and inflation



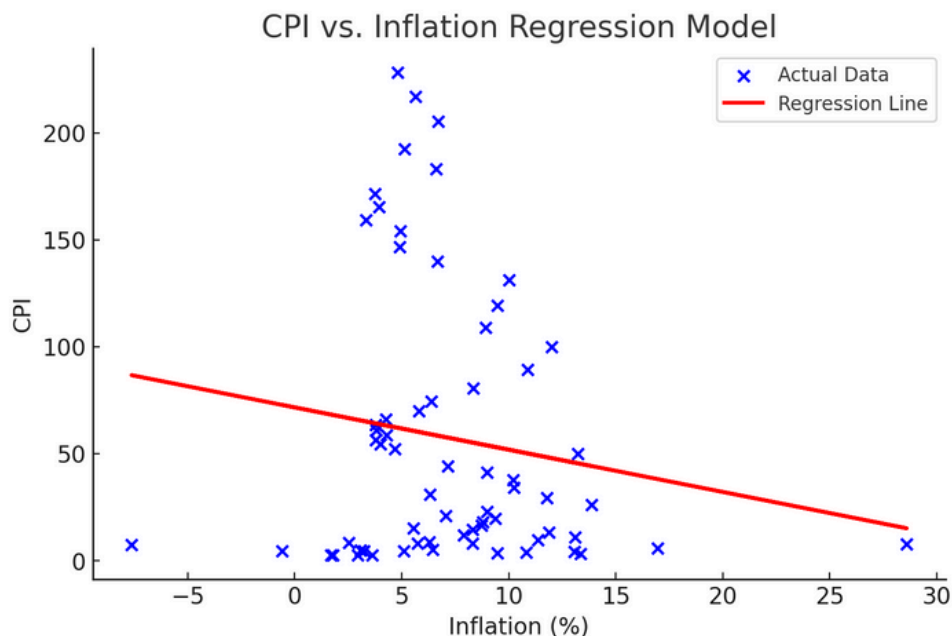
- CPI (Blue Line) shows a steady increase over time, indicating long-term inflation.
- Inflation Rate (Red Dashed Line) fluctuates, with spikes in the 1970s & early 2010s, followed by periods of relative stability.
- Despite fluctuating inflation, CPI continues to rise, meaning that prices rarely decrease even when inflation drops.

# Inflation Volatility Analysis



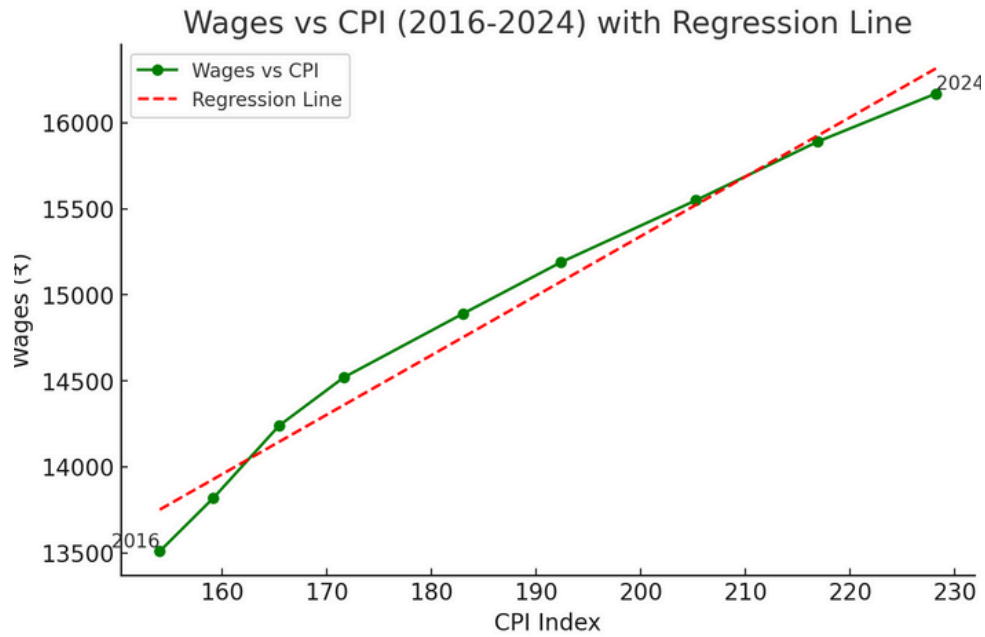
- The red line shows actual inflation values, highlighting high volatility over the years.
- The blue dashed line (5-year moving average) smooths out short-term fluctuations and shows long-term trends.
- Extreme spikes occurred in the 1970s (oil crisis), 1990s, and post-2008 periods.
- Since 2000, inflation has been more stable, but there are still fluctuations.

## Regression Model (CPI and Inflation)



- CPI steadily increases over time, while inflation fluctuates yearly, which explains the weak correlation.
- Other economic factors (interest rates, wages, demand shocks, etc.) might play a bigger role in CPI growth.

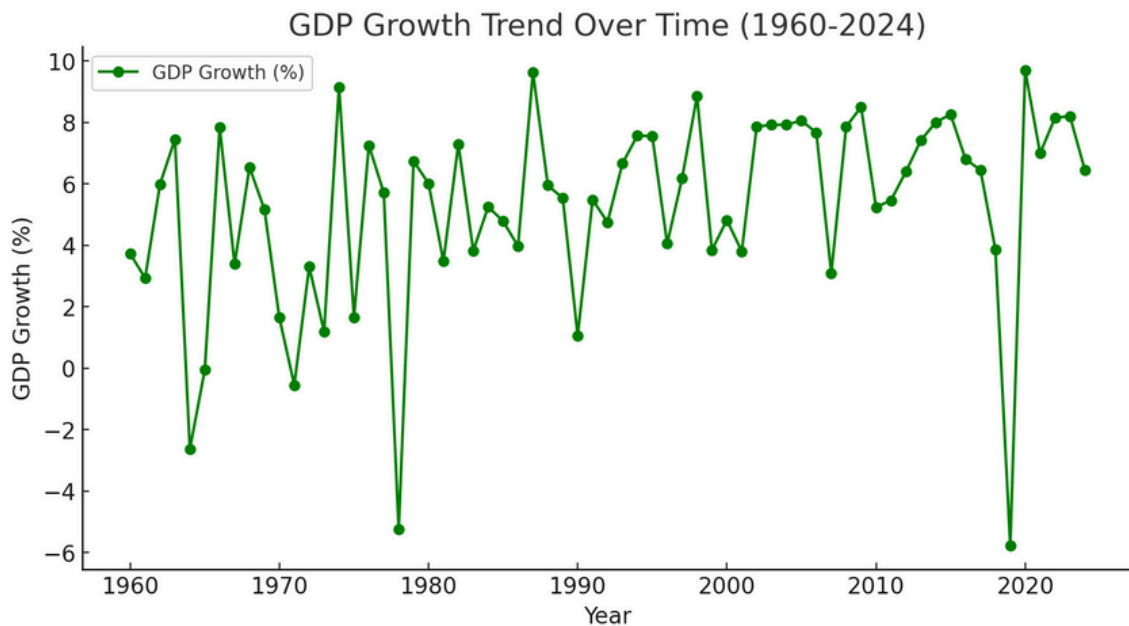
# CPI vs Wage



- Green Line (Actual Data): Wages have steadily increased as CPI rises.
  - Red Dashed Line (Regression Line): The close alignment suggests a strong positive correlation—wages and CPI move together.
  - Any deviation from the trend could indicate periods where wages fell behind or outpaced inflation.
  - Purchasing Power: If the wage line is above the regression line, wages are growing faster than inflation → Better purchasing power.
  - If it's below, wages aren't keeping up with CPI → Reduced real income.
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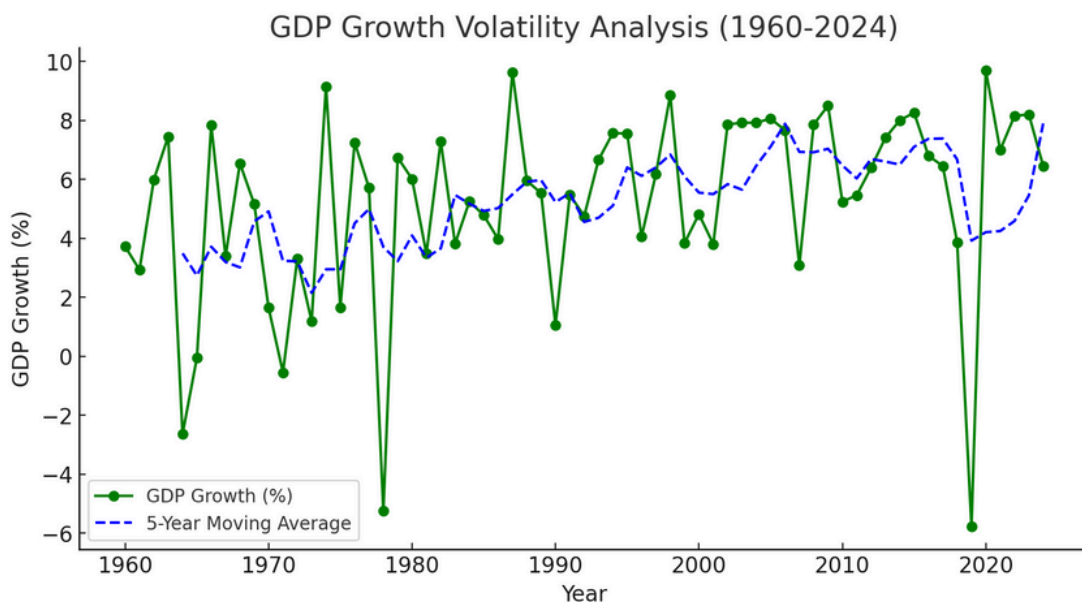


# GDP Growth



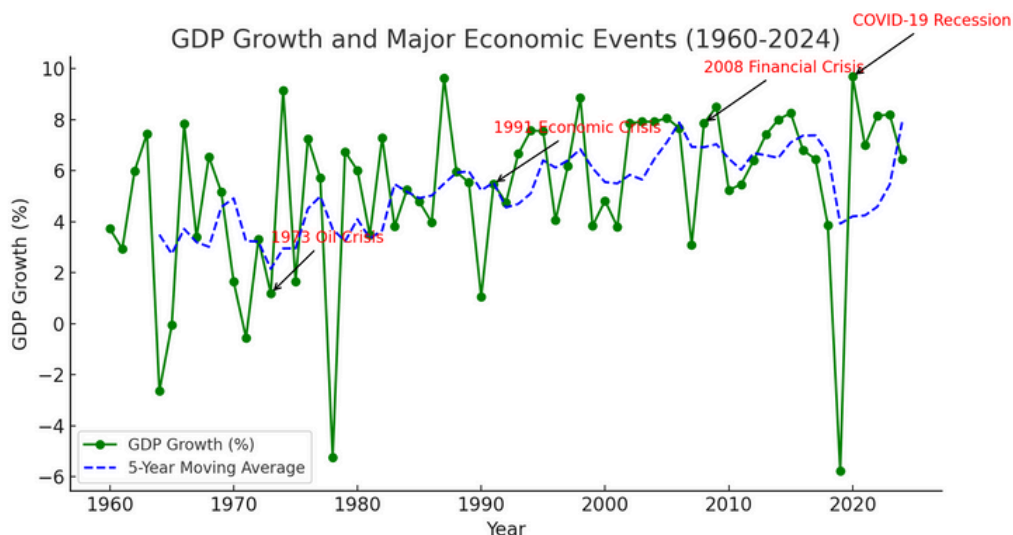
- GDP growth is highly volatile, with several sharp downturns (e.g., early 1970s, late 2000s, and 2020).
- Strong economic expansions occurred in the late 1980s, early 2000s, and post-2020.
- The 2020 crash is the steepest, likely due to the COVID-19 pandemic.

## GDP Growth Volatility



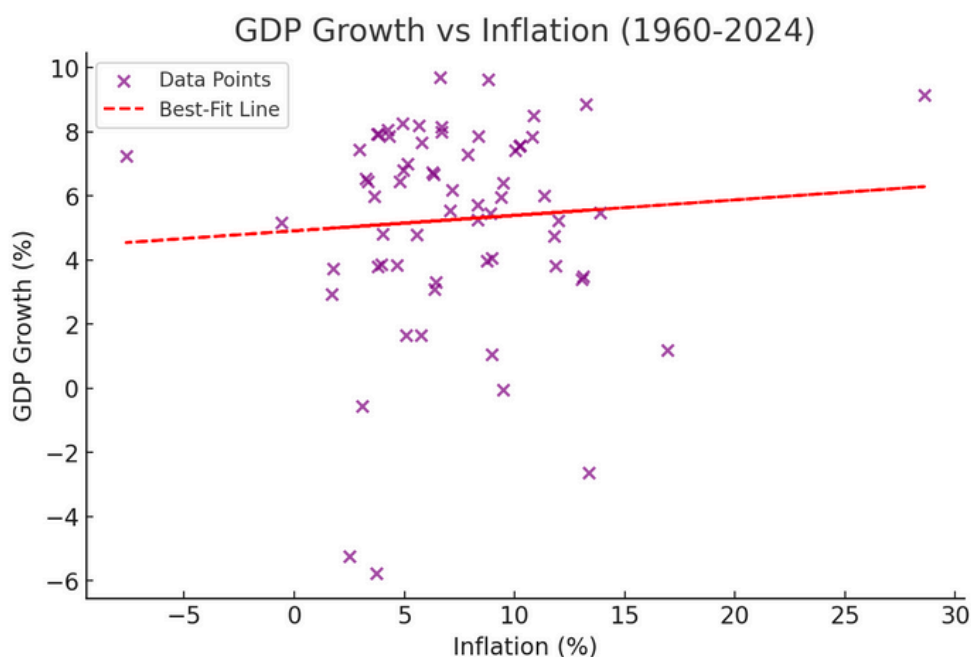
- Green Line → Raw GDP growth values show high volatility, with extreme fluctuations in some years.
- Blue Dashed Line (5-Year Moving Average) → Smooths short-term variations, revealing long-term economic trends.
- Major dips occur during economic crises (e.g., 1970s oil crisis, 2008 financial crash, and 2020 pandemic).
- Recent GDP trends appear more stable, though there are still occasional downturns.

# GDP Growth and Major Economic Events



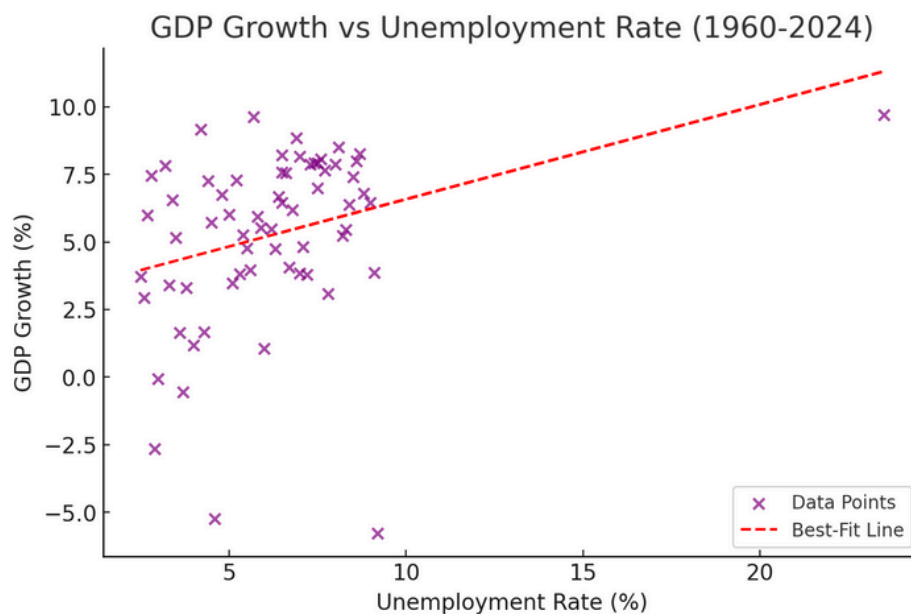
- Major downturns in GDP growth align with global economic crises: 1973 Oil Crisis → First major shock causing stagflation.
- 1991 Economic Crisis → Marked India's economic liberalization era.
- 2008 Financial Crisis → Global recession, causing GDP slowdown.
- 2020 COVID-19 Recession → Steepest drop due to lockdowns and economic shutdown.
- The blue dashed line (5-year moving average) smooths out volatility, showing underlying trends.

## GDP Growth vs Inflation



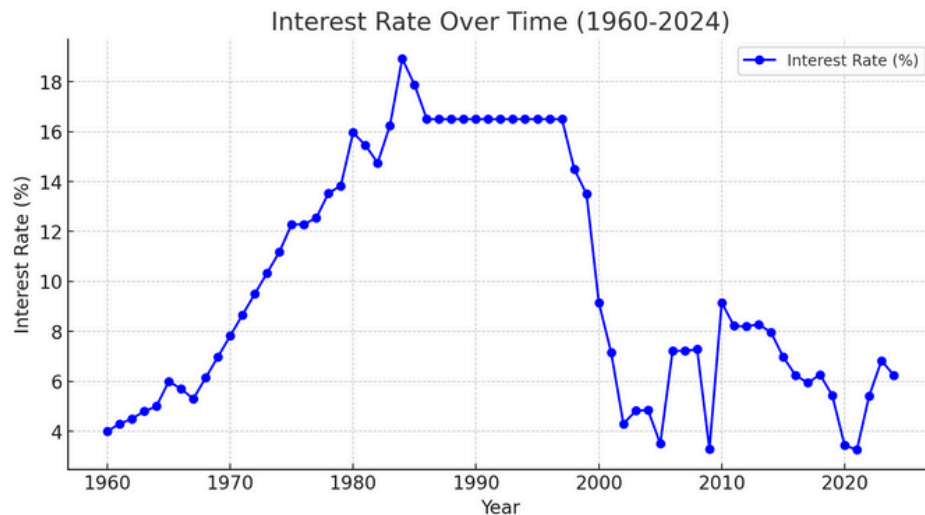
- Purple Dots: Each point represents a specific year's GDP growth and inflation rate.
- Red Dashed Line (Best-Fit Line): The downward slope suggests a negative relationship between inflation and GDP growth.
- High inflation may slow economic growth, though some periods show exceptions.
- Clusters & Trends: Low inflation (<5%) tends to have mixed GDP growth—some strong, some weak.
- Periods of high inflation (>10%) often coincide with lower GDP growth or recessions.

# GDP Growth vs Unemployment



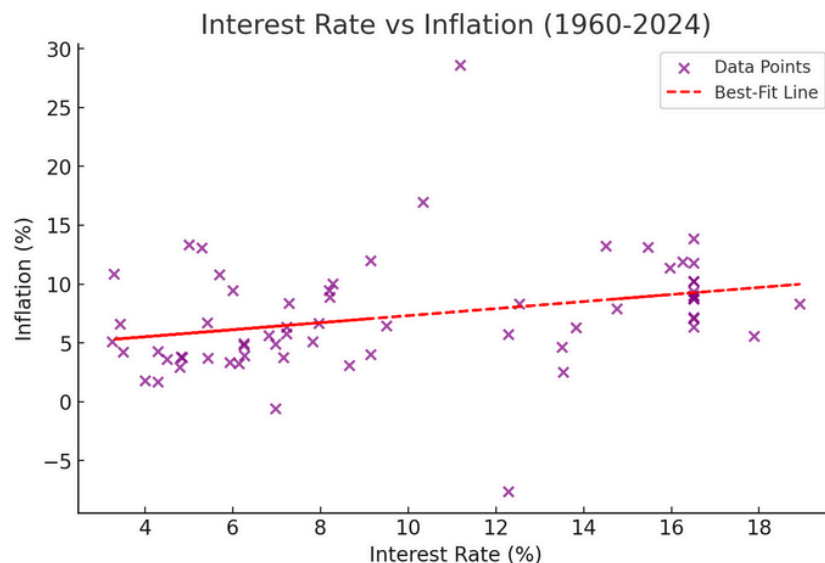
- Purple Dots: Each point represents a specific year's GDP growth and unemployment rate.
  - Red Dashed Line (Best-Fit Line): The slightly positive slope is unexpected—typically, we expect a negative correlation (higher unemployment → lower GDP growth).
  - However, some outliers exist (e.g., 2020 pandemic, where GDP dropped sharply but unemployment spiked).
  - Clusters & Trends: Moderate Unemployment (4-8%) is common, with GDP growth fluctuating around 3-7%.
  - High Unemployment (>9%) is rare but usually coincides with economic crises.
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# Interest



- Steady rise in interest rates from the 1960s to early 1980s, likely due to inflationary pressures.
- Peak interest rates (~18-19%) in the 1980s, possibly linked to global economic policies.
- Sharp decline in the late 1990s and early 2000s, indicating monetary easing.
- Fluctuations post-2008, reflecting responses to the financial crisis and pandemic recovery.

## Interest Rate vs Inflation



- Purple Dots: Each point represents a specific year's interest rate and inflation.
- Red Dashed Line (Best-Fit Line): The slight upward trend suggests a weak positive correlation, meaning higher interest rates did not always reduce inflation effectively.
- This could indicate periods of stagflation (high inflation + high interest rates).

### Clusters & Outliers

- Some high inflation years still had high interest rates, showing that monetary policy wasn't always successful.
- There are some negative inflation points (deflation), where interest rates were relatively low.

# Co-Relation Analysis

## CPI vs. Inflation → Weak Correlation

- Observation: CPI steadily increases over time, while inflation fluctuates significantly.
- Analysis: A weak correlation suggests that inflation alone does not fully drive CPI growth. Other economic factors like interest rates, wages, and supply shocks also contribute.

## CPI vs. Wages → Strong Positive Correlation

- Observation: Wages and CPI move together, meaning as inflation rises, wages generally increase too.
- Analysis: This is expected, as wages typically adjust for inflation. However, periods where wages lag behind CPI growth indicate a decline in real income and purchasing power.

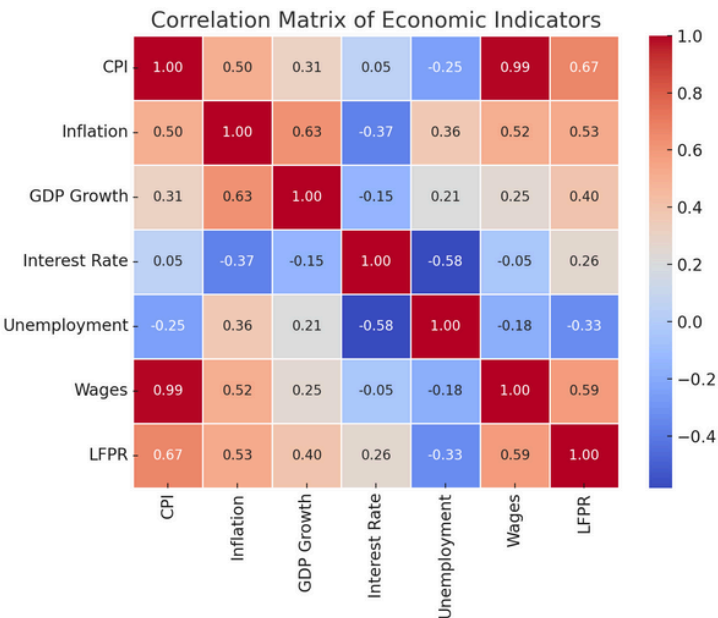
## GDP Growth vs. Inflation (Moderate Negative Correlation)

- Observation: When inflation is very high (>10%), GDP growth is often lower.
- Analysis: This suggests that high inflation may slow economic growth—a sign of stagflation. However, there are exceptions (some low-inflation periods with weak GDP growth).

## GDP Growth vs. Unemployment (Weak or Unexpected Correlation)

- Observation: There is no clear inverse correlation—sometimes GDP shrinks but unemployment remains low (e.g., 2020 COVID crash).
- Analysis: This could be due to government interventions like stimulus packages or differences in labor market conditions.

# Summary



This heatmap shows how key economic indicators relate to each other. Stronger correlations indicate a direct relationship, while weaker ones suggest external factors play a bigger role.

## Key Findings:

### Strong Correlations:

- CPI & Wages (0.99): Wages rise with inflation, but this doesn't confirm if real wages are improving. A real wage analysis is needed.
- Inflation & GDP Growth (0.63): Inflation tends to rise during economic booms, but excessive inflation can slow growth.
- LFPR & CPI (0.67): Higher workforce participation is linked to increased inflation, possibly due to higher demand and wage pressures.

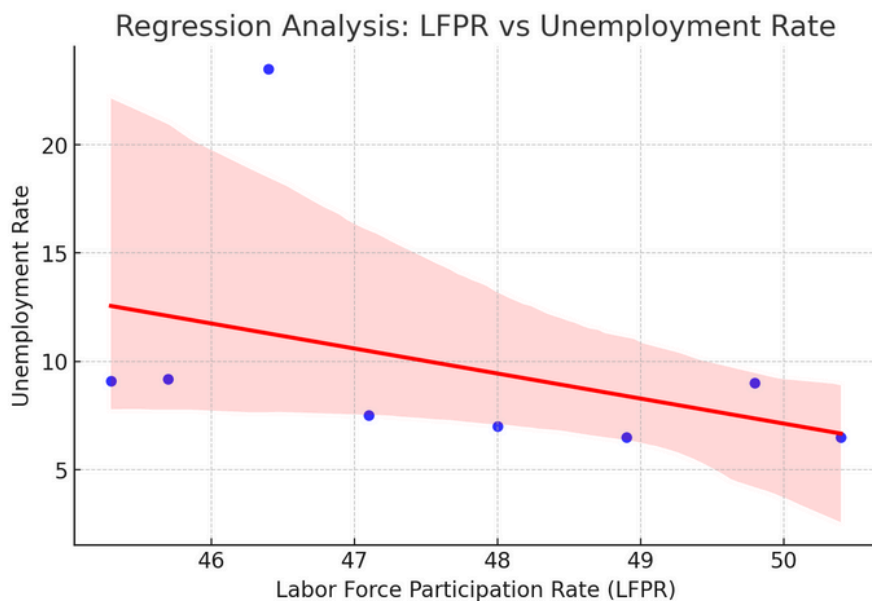
### Moderate Correlations:

- LFPR & Wages (0.59): A larger workforce may push wages up, but other factors like job availability and productivity matter.
- Inflation & Wages (0.52): Wages adjust to inflation, but real income stability depends on whether wages rise faster than CPI.
- GDP Growth & LFPR (0.40): Economic growth may encourage workforce participation, but structural labor issues can weaken this link.

### Weak or Negative Correlations:

- Interest Rate & Inflation (-0.37): Raising interest rates hasn't always controlled inflation, suggesting policy delays or external price shocks.
  - Unemployment & LFPR (-0.33): More people entering the workforce doesn't guarantee lower unemployment, possibly due to job market mismatches.
  - Unemployment & Wages (-0.18): No clear Phillips Curve effect—suggesting wage growth is driven by other factors like automation, industry shifts, or policy changes.
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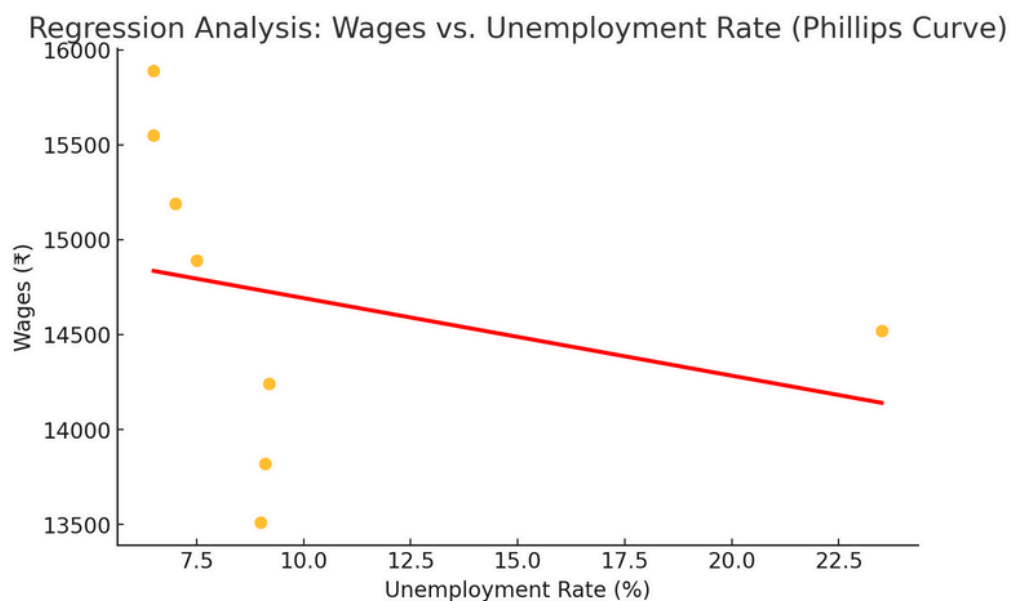
# LFPR vs Unemployment : Regression Model



## Key Findings from the Regression Model:

- Negative Relationship:** The coefficient for LFPR (-1.1558) suggests that as labor force participation increases, the unemployment rate tends to decrease. However, this relationship is not statistically significant.
- Low R-Squared (0.149):** Only 14.9% of the variation in the unemployment rate is explained by LFPR, meaning other factors heavily influence unemployment.
- P-Value (0.344):** The relationship is not statistically significant ( $p > 0.05$ ), implying that we cannot confidently say that LFPR directly impacts unemployment.
- Outlier Effect (2020 Spike):** The unemployment spike in 2020 (COVID-19 impact) might distort the trend.

## Wages vs. Unemployment Rate (Phillips Curve Relationship)



continued...

- Negative Relationship (-40.89 Coefficient):
- The coefficient is -40.89, meaning that for every 1% increase in unemployment, wages decrease by ₹40.89 on average.
- This aligns with Phillips Curve theory, which suggests wages tend to decline when unemployment is high.
- Low R-Squared (0.077 or 7.7%):
- Only 7.7% of wage variations are explained by unemployment, suggesting other factors like inflation, productivity, or policy changes impact wages more.
- High P-Value (0.505) → Not Statistically Significant:
- Since  $p > 0.05$ , we cannot confidently say that unemployment significantly impacts wages in this dataset.
- The weak relationship might be due to external shocks (COVID-19 in 2020, economic policy shifts, etc.).

Economic trends like inflation, unemployment, wages, and GDP growth are shaped by policy changes, automation, and global shocks beyond market forces.

### Policy Changes:

- Interest rate hikes don't always control inflation immediately due to policy delays.
- Fiscal policies (subsidy, tax cuts) boost GDP growth but may increase inflation.
- Labor laws (minimum wage hikes) raise wages, breaking the expected unemployment-wage link.

### Automation & Tech Shifts:

- Automation eliminates low-skill jobs, weakening the Phillips Curve.
- Productivity gains increase profits without raising wages.
- Industry shifts (labor intensive manufacturing to tech driven production) change how GDP growth affects employment.

### Pandemics & Global Shocks:

- COVID-19 caused record high unemployment (23.5%) unrelated to economic weakness.
  - Supply chain disruptions drove inflation, even when demand fell.
  - Governments cut interest rates for recovery, later causing inflation spikes.
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# Conclusion

- This analysis of India's economic landscape reveals a tapestry of interconnected yet fluid relationships. While CPI and wages march upward in tandem, true purchasing power hinges on whether wages outpace inflation—a race where workers often struggle to stay ahead. Inflation and GDP growth, though occasionally allies, turn adversarial when unchecked price rises stifle long-term prosperity, echoing the delicate balance policymakers must strike.
  - Surprisingly, unemployment and wages defy conventional wisdom: automation, labor reforms, and sectoral shifts (like tech's dominance) often overshadow job availability in shaping income trends. Similarly, interest rates, while a trusted tool for central banks, falter in taming inflation during structural crises—a reminder that economies resist simple fixes.
  - External forces loom large here. The COVID-19 pandemic, globalization, and India's economic reforms since 1991 act as plot twists, reshaping trends overnight. These dynamics underscore a truth: economies are living ecosystems, not static models.
  - For policymakers and analysts, this means embracing adaptability. Prioritizing real wage growth, balancing inflation without stifling innovation, and crafting labor policies for a digitizing world are critical. For citizens, it highlights the importance of financial literacy in navigating an unpredictable landscape.
  - In the end, economics is less about rigid formulas and more about navigating uncertainty. As India evolves, so too must our strategies—ready for the next crisis, opportunity, or revolution hiding just beyond the horizon.
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